`canyon-0.2.1` Manual

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https://github.com/cnaak/canyon.typ
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1 Canyon's End User in Full Control Approach

Mainstream Typst templates are written so as to bring functionality and to be used in an `#import` then `#show` workflow; however, that frequently comes at the cost of introducing (i) new syntax and (ii) opaque template-side code and settings.

Any template end user that had the need to customize things not anticipated or provided for by the template's author have experienced this shortcoming.

Canyon (`canyon-0.2.1`) explores a different approach in which a new canyon-based document is a book *stencil* that users can freely edit and adapt, with:

- *No* opaque template-side code or settings;
- *No* canyon `#import` statements;
- *No* canyon-specialized `#show` rules;
- *Full access* to the entire Typst project;
- The *familiar* Typst syntax to work with; and
- Full *separation of concepts* for (i) metadata, (ii) configuration, (iii) elements, (iv) layout, and (v) contents.

Therefore, canyon users have access to *all* document settings as if they had manually started writing the document from scratch on their own—except they hadn't!—and thus have *full access* to everything Typst in that project.

2 Overview

Out-of-the-box canyon books come with the following:

1. (i) Front-matter; (ii) Body; and (iii) Back-matter book structure;

2. (i) Cover page, and (ii) numbered Chapter page images;
3. All `#set` and `#show` rules are accessible; hence, customizable;
4. The "openright" chapter pagination—through `#pagebreak(to: "odd")`;
5. Front-matter with Title pages; Credits; Contents; List of Figures, Tables, Equations, and Exhibits; and a Preface;
6. Book body with regular Chapters (single Part);
7. Back-matter with Appendix, Bibliography, Glossary, Index, and Disclaimers;
8. Figures, Tables, Exhibits, Problems, and Equations are Chapter-numbered: `1.1` instead of `1`, with configurable separators by type *group*, i.e., a separate one for: (i) Equations, (ii) Figures and Tables, (iii) Exhibits, and (iv) Problems;
9. A simple and configurable 3-color system with: (i) a thematic color (default: *dark indigo*), (ii) a page background color (default: *light sepia*), and (iii) a cover text color (default: *dark indigo*).

A *selection* of canyon book pages using default settings are grouped in the thumbnail below:

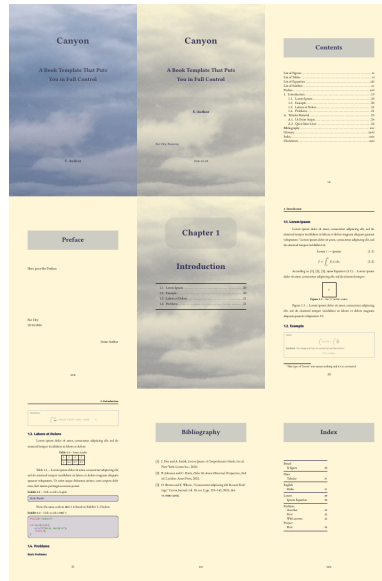


Figure 1: Ordered subset of canyon pages: sample (i) front-matter: half-title, title, contents, and preface; (ii) book body; and (iii) back-matter pages

2.1 Stencil Overview

The stencil tree delivered to the end user upon template instantiation—either through creating a new document from template in `typst.app`, or through the `typst init` command line on a shell—is shown on Listing 1:

```
./
|-- 1-FRONT/
|-- 2-BODY/
|-- 3-BACK/
|-- canyon.pdf
|-- canyon.typ
|-- RES/
`-- SETUP/
```

Listing 1: Canyon stencil level-1 tree. Entries followed by a slash are directories

The numbered `1-FRONT`, `2-BODY`, and `3-BACK` directories are for (i) front, (ii) body, and (iii) back-matter book sections—the numbering is to maintain sorting order, specifically on `typst.app` files environvent—the `RES` dir is for document resource files—figures, bibliography, etc.—, and the `SETUP` dir is for document-wide metadata and settings, as well as for 3rd-party package settings.

3 Suggested Canyon Workflows (Quick Start)

End users are free to adopt any book-writing workflow as they see fit (or even adapt the stencil to something else, as a manual); however, the following ones are suggested as a quick start, for their simplicity, shallow learning curve, and perceived effectivity:

3.1 The Basic Workflow: Metadata and Contents

- Quickly editing `SETUP/META.typ` for document *metadata*;
- Setting up your actual cover/chapter art `RES/<images>`;
- Adding custom user Typst commands on `SETUP/USER.typ`;
- Being up and running for *content* `2-BODY/<chapters>` creation and adaptation;
- Simply adjust book front- (`1-FRONT/<sections>`) and back- (`3-BACK/<sections>`) matter *contents*.

The scope of the `META` dictionary defined in `SETUP/META.typ` is the entire book, and includes, as of `canyon-0.2.1` the `title`, `subtitle`, `author`, `auth`, `address`, `description`, `date`, and `keywords` entries.

This workflow end-result will look like Figure 1 except with your personalized artwork, metadata, contents, and general front- and back-matter portions.

3.2 The Configuration Workflow: General Settings

- All the Basic workflow with:
- Quickly setting *general document configurations* on `SETUP/CONFIG.typ`;
- Editing `__setup.typ` files in the `1-FRONT`, `2-BODY`, and `3-BACK` template stencil directories to more finely control how the corresponding *broad book* sections are displayed.

The scope of the `CONFIG` dictionary defined in `SETUP/CONFIG.typ` is *main book settings*, and includes, as of `canyon-0.2.1` the main `lang`, `paper`, `size`, `fonts`, `colors`, and `num-seps`, i.e., reference chapter-to-item number separarators, as the `\u{2013}` in "Problem 1–1", for instance.

This workflow end-result improves upon the Basic one by setting custom language, paper, book matter sections, text size, fonts, and broad color scheme.

3.3 The Fine-Grained Workflow: Element and Layout

- All the Configuration workflow with:
- Editing `SETUP/ELEMENTS.typ` for fine-grained element *customizations* such as: (i) the set of typst native `page`, `par`, `text`, and `block` elements; and (ii) custom `canyon` elemens, such as: `exhibit` (for code listings, examples, etc.), and `problem` (for textbooks); or even (iii) creating your own custom element types;

The scope of the `ELEM` dictionary defined in `SETUP/ELEMENTS.typ` is *various pre-defined book elements*, and includes, as of `canyon-0.2.1`, native Typst *elements* such as: `page`, `par`, `text`, and `block`—meaning a *plurality* of *custom* {`page`, `par`, `text`, and `block`}-setting definitions.

These were designed to be *fully generic*, thus accepting *all valid* named parameter definitions as these native Typst elements can accept—not just a subset thereof—owing to the way this information is used in conjunction with Typst's spreading (`..`) operator on dictionaries as illustrated on Listing 2.

This workflow end-result improves upon the Configuration by allowing for *a highly customized book*, than can bear little to no resemblance to the default canyon book style, hopefully achieved without too much pain for the end user.

In `SETUP/ELEMENTS.typ` `canyon-0.2.1` defines, among other variants, the following `ELEM.page` entries as a dictionaries of valid Typst `page` function named arguments:

```
// blank page
#ELEM.page.insert("blank", (
  paper: CONFIG.paper,
  margin: margins.graphic,
  header: none,
  background: none,
  fill: none,
  numbering: none,
))

  // half-title (book cover) page
#ELEM.page.insert("half-title", ELEM.page.blank + (
  background: image(cover, width: 100%),
))
```

This means that `ELEM.page.half-title` "inherits" (builds upon) `ELEM.page.blank` with a overridden (custom) `background` setting.

This definition allows it to be used as:

```
#page(..ELEM.page.half-title)[
  ...
]
```

in `1-FRONT/half-title.typ` for actually producing the half-title page.

Canyon element definitions are programatically *expressive*—as `half-title` page is just the `blank` page with a custom `background`—while maintaining full flexibility—as nothing prevents an end user to add any or all of the valid Typst `page` named arguments to it, such as `binding` or `columns`, etc., according to Typst's documentation.

Listing 2: Sample canyon element definition and usage design, highlighting its expressiveness and full versatility

4 Citing

This package can be cited with the following bibliography database entry:

```yaml
canyon-package:
  type: web
  author: Naaktgeboren, C.
  title: "Canyon: A Book Template That Puts You in Full Control"
  url: https://github.com/cnaak/canyon.typ
  version: 0.2.1
  date: 2026-10-29
```